

Control Systems (EEE/ECE/INSTR F242)

Tutorial 3 (27 January 2026)

1. For the block diagram shown in Fig. Q1(a) and Q1(b), first draw the signal flow graph, and then using Mason's gain formula, find closed loop transfer function $G(s) = C(s)/R(s)$.

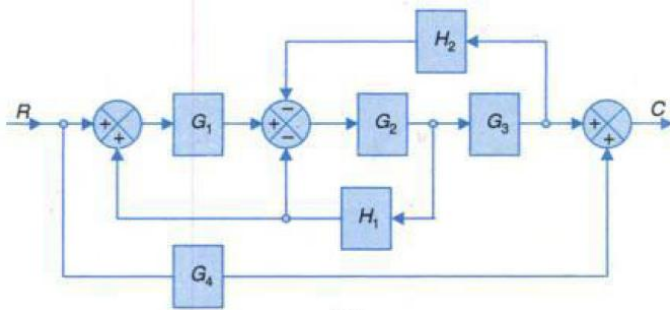


Fig. Q1(a)

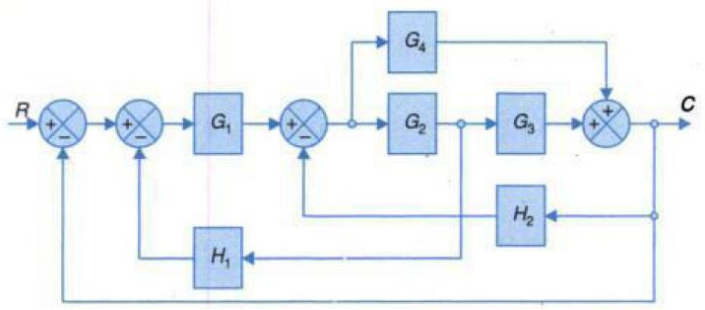


Fig. Q1(b)

2. For the liquid level system of two interacting tanks shown in Fig. Q2, the governing equations can be written as follows. Draw the signal flow graph, and then using Mason's gain formula, find closed loop transfer function $G(s) = \Delta Q_2(s)/\Delta Q(s)$.

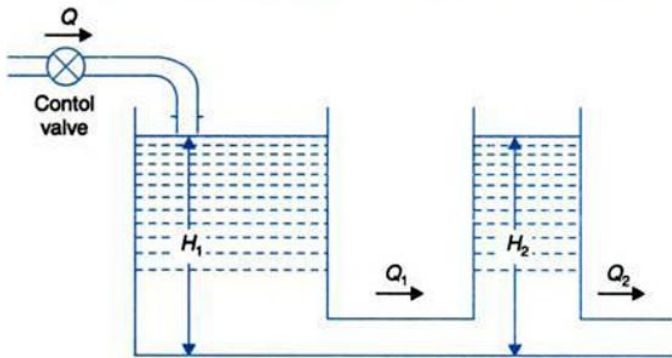


Fig. Q2

$$sC_1\Delta H_1(s) = \Delta Q(s) - \Delta Q_1(s)$$

$$\Delta Q_1(s) = (\Delta H_1(s) - \Delta H_2(s))/R_1$$

$$sC_2\Delta H_2(s) = \Delta Q_1(s) - \Delta Q_2(s)$$

$$\Delta Q_2(s) = \Delta H_2(s)/R_2$$

Where, C_1, C_2 are the capacity of tanks 1 and 2, respectively. R_1, R_2 are the hydraulic resistances for the output flow of tanks 1 and 2, respectively. $\Delta H_1, \Delta H_2, \Delta Q_1$, and ΔQ_2 are the deviation variables from the steady states of H_1, H_2, Q_1 and Q_2 , respectively.
